Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	XMMUM
Project / Solution Name	pravista
Name (Leader)	TAN ZHI KAI
Email & Phone (Leader)	kai64327174@gmail.com , 011-6432 7174
University Affiliation(s)	XMUM, MMU
Live Prototype URL	
Code Repository URL (public / access granted to judges)	

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Pravista is structured as an asynchronous web application with multiple tiers engineered for zero-trust privacy, as well as high responsiveness:

1. **Client Tier**: Built with Next.js 16, styled with Tailwind CSS and GSAP ScrollTrigger and Flip animations. Client state and caching are coordinated by TanStack React Query v5, with client-side document rendering handled by PDF.js.
2. **API and Business Logic**: An asynchronous FastAPI backend executes deterministic career suitability criteria and ephemeral CV extraction using PyMuPDF and Tesseract OCR. Raw uploaded documents are then discarded immediately after in-memory parsing.
3. **AI services**: Bounded external LLM calls are executed strictly at the server side using HTTPX with structured schema validation, without exposing the key to the client.
4. **Database and Cloud Storage**: Supabase PostgreSQL manages rational persistence across 17 tables protected by native RLS and JWT verification via PyJWT. The uploaded course materials are stored in private Supabase Storage buckets, and are delivered to verified enrolled users through temporary signed bearer URLs.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
@fontsource-variable/geist 5.3.0	Local Geist font	OFL-1.1
@fontsource-variable/inter 5.3.0	Local Inter font	OFL-1.1
@hookform/resolvers 5.9.2	Form validation adapter	MIT
@radix-ui/react-slot 1.5.0	UI composition	MIT
@supabase/supabase-js 2.118.0	Authentication client	MIT

Name & Version	Purpose in Project	License
@tanstack/react-query 5.104.5	API state and caching	MIT
@xyflow/react 12.12.0	Interactive impact graph	MIT
class-variance-authority 0.7.1	Component variants	Apache-2.0
clsx 2.1.1	CSS class composition	MIT
gsap 3.15.0	Interface animation	GSAP standard no charge license
lucide-react 0.468.0	Interface icons	ISC
motion 12.43.0	Interface animation	MIT
next 16.4.2	Web framework	MIT
pdfjs-dist 6.4.299	CV PDF preview	Apache-2.0
react 19.3.1	Interface components	MIT
react-dom 19.3.1	Browser rendering	MIT
react-hook-form 7.90.0	Form state	MIT
recharts 3.10.1	Charts	MIT
tailwind-merge 3.7.0	CSS utility merging	MIT
zod 4.6.7	Input schemas	MIT
@playwright/test 1.64.1	Browser tests	Apache-2.0
@tailwindcss/postcss 4.3.3	CSS build integration	MIT
@testing-library/jest-dom 6.9.1	DOM assertions	MIT
@testing-library/react 16.3.3	Component tests	MIT
@types/node 24.19.2	Node type definitions	MIT
@types/react 19.3.0	React type definitions	MIT

Name & Version	Purpose in Project	License
@types/react-dom 19.3.0	React DOM type definitions	MIT
jsdom 26.1.0	Test browser environment	MIT
tailwindcss 4.3.3	CSS utilities	MIT
typescript 5.9.3	Type checking	Apache-2.0
vitest 5.0.5	Component and unit tests	MIT
aiosqlite 0.22.1	Isolated test database	MIT
alembic 1.20.0	Migrations	MIT
asyncpg 0.32.0	PostgreSQL driver	Apache-2.0
fastapi 0.155.0	HTTP API	MIT
httpx 0.28.1	External HTTP calls	BSD-3-Clause
pydantic 2.14.1	Data validation	MIT
pydantic-settings 2.16.1	Configuration	MIT
PyJWT 2.15.1	JWT verification	MIT
pymupdf 1.28.2	PDF parsing and preview	GNU Affero GPL 3.0 or Artifex commercial license
pytest 9.1.1	API and engine tests	MIT
pytest-asyncio 1.4.0	Async test support	Apache-2.0
python-docx 1.2.0	DOCX extraction	MIT
python-multipart 0.0.32	Upload handling	Apache-2.0
ruff 0.17.1	Code checks	MIT
SQLAlchemy 2.1.5	Database access	MIT

Name & Version	Purpose in Project	License
uvicorn 0.54.0	ASGI server	BSD-3-Clause
Python >=3.12; JavaScript; SQL	Backend and scripts; browser code; database queries	Language requirements; library licenses listed above
Tesseract; LibreOffice (optional; versions not pinned)	Local OCR and DOCX conversion paths	Apache-2.0
Node.js 22.13+	Frontend runtime environment	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Supabase Auth API	User registration, email verification, and ES256 JSON Web Key Set (JWKS) token validation	Free
OpenAI-Compatible Chat Completions API	Optional CV writing suggestions and formatting improvements using strict JSON schema enforcement	Free
Google Gemini API	Semantic evaluation of student project logs to determine UN SDG target allocations and contextual role summaries	Free
OpenWeb Ninja	Programmatically search, retrieve, and filter millions of job vacancies from Jooble's global job board	Free
Google News RSS Search Feed	Fetching real-time global and Malaysian SDG news headlines matched by user skill keyword	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Supabase PostgreSQL and Auth integration	Relational database (17 tables with RLS for profiles, plans, courses, SDG trackers) and authentication	Free
Local development processes	Frontend Next.js (port 3000) and FastAPI Uvicorn ASGI server (port 8000) runtime environments	Local configuration

Service & Provider	Purpose in Project	Plan / Region
Supabase Storage	Private and secure cloud storage for uploaded hosted-learning lesson media. Access via 6-hour signed URLs.	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Tesseract OCR	Local executable	Pre-trained (Scanned page and image OCR; document_layout.py)	Apache 2.0
Gemini 1.5 Flash	Google AI Studio	Pre-trained (Zero-shot semantic assessment with strict system prompt and Pydantic JSON schema output)	Google AI Terms of Service
OpenAI GPT-4o-mini	OpenAI	Pre-trained (Structured JSON schema suggestions for CV wording review)	Commercial API Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
WEF Future of Jobs 2025	weforum.org	Public	Open Citation Reference	N
O*NET Occupational Taxonomy	onetonline.org	Public	Creative Commons Attribution 4.0 (CC BY 4.0)	N
UN SDG Targets & Indicators	sdgs.un.org	Public	UN Citation	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Curated Skill & Career Catalogue	backend/app/catalog.py (Adapted from ESCO/O*NET)	Synthetic	Self-created	N
Fictional Persona Records	backend/app/demo.py and frontend/src/lib/demo-preview.ts	Synthetic	Self-created	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✅ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude 3.7	Code scaffolding, test case generation, and refactoring	Moderate
Gemini 3.8 Flash	Code scaffolding, test case generation, and refactoring	Moderate
GitHub Copilot	Boilerplate syntax auto-completion in TypeScript and Python	Minor
ChatGPT (GPT-4o)	Drafting mock CV text samples for test fixtures	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team used generative AI assistants for initial boilerplate generation, API scheme mapping, and test fixture scaffolding. All generated codes were inspected manually, refactored, and tested against strict type definitions and security standards. We implemented a comprehensive verification suite featuring 50+ frontend component tests, 30+ backend integration tests, and Playwright end-to-end workflows. Core career-matching criteria and scoring algorithms were independently implemented using deterministic logic to ensure predictable and reproducible results. Our team assumes full technical responsibility for the architecture and implementation

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✅ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Development Credits	Used as an auxiliary prototyping environment
Replit	URL	Upload as URL
Runpod	API	Used as a remote platform for AI running

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was utilized as a supporting tool for early brainstorming and development of prototype. However, it was not essential to the core system functionality and has been replaced. The final prototype operates independently on Next.js and FastAPI stack connected to Supabase cloud infrastructure. The application can be built, tested, and deployed on standard cloud platforms or local environments using our automated setup scripts.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The platform enforce strict data minimisation and zero-trust privacy

1. **Ephemeral Document Processing**: All uploaded CVs are parsed entirely in transient memory using PyMuPDF and Tesseract OCR. Raw files are discarded immediately. No original documents are retained on disk or cloud
2. **PII Redaction**: Optional AI wording suggestions explicitly strip candidate name, email, and phone numbers before transmission of API
3. **Database Isolation**: Supabase PostgreSQL enforces Row-level Security (RLS) across all the public tables. Unauthenticated access is revoked, and authenticated users can only access their own records through ES256 JWT validation
4. **Private Storage**: Course lesson media files are stored in private Supabase Storage buckets and accessed exclusively through a signed URLs that are available for 6 hours only after verifying course enrollment
5. **Data Retention**: Users retain full control over their data. When a CV is updated, outdated assessments and career pathways are automatically removed

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. **AI Hallucinations**: Generative AI can invent skills or qualifications
   Mitigation: We completely decouple career scoring with generative AI. All fit percentages, rubric breakdowns, and skill comparisons are done using a deterministic Python algorithm.
2. **External API Latency and Outages**: Failure in external services could interrupt user experience
   Mitigation: The backend HTTPX client uses 12 to 15 second timeouts, automatic retry, and predefined the fallback rules when external services failed. The demo also includes an offline data package to keep essential features available during network outages
3. **OCR Variances**: Scanning CV with low resolution may cause inaccurate data extraction
   Mitigation: The UI provides an interactive verification panel highlights fields with confidence scores below 85%, printing users to review and verify the information

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

✅ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

✅ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

✅ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

✅ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

✅ Any use of official SDC sponsor resources has been declared in Section 5.

✅ The prototype described here is accessible to judges through the links provided in Section 1.

✅ All fields in this template have been completed within the word limits stated.

8. Declaration

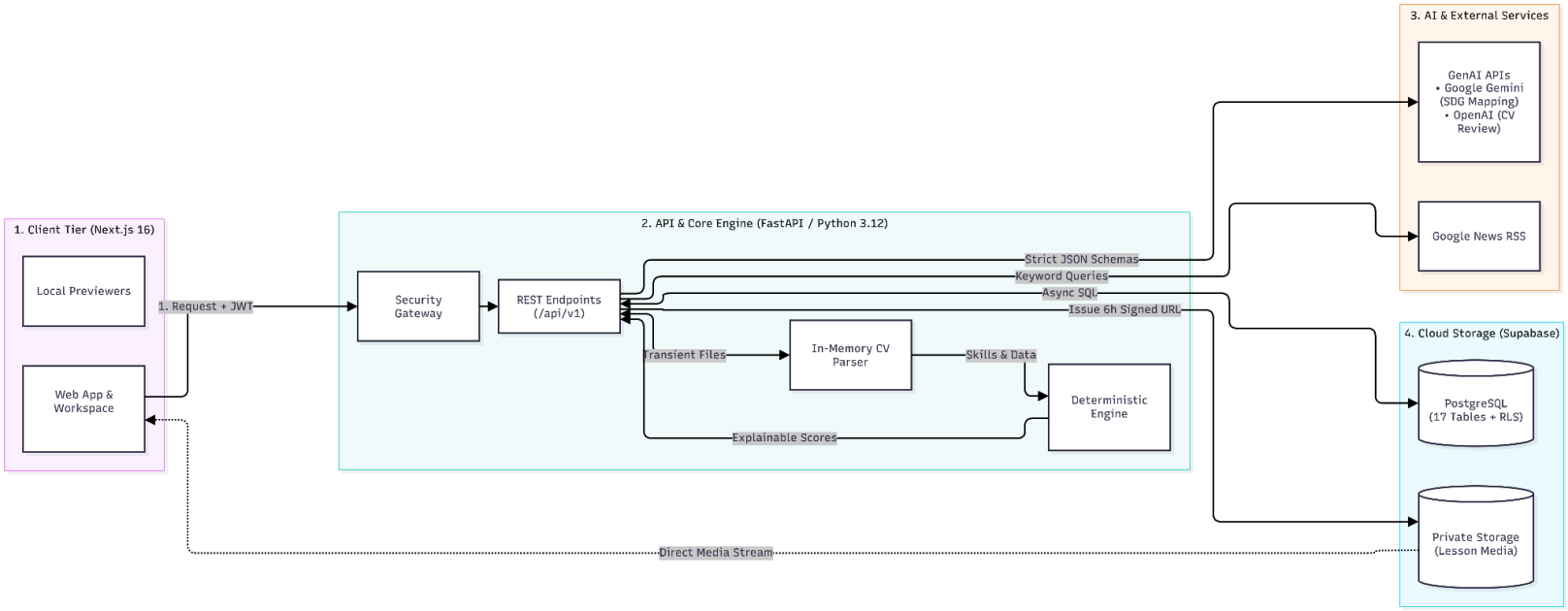
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[25/10/2026]

Appendix


1. Client Tier (Next.js 16)
Local Previewers
Web App & Workspace
1. Request + JWT
2. API & Core Engine (FastAPI / Python 3.12)
Security Gateway
REST Endpoints (/api/v1)
Transient Files
In-Memory CV Parser
Skills & Data
Deterministic Engine
Explainable Scores
Strict JSON Schemas
Keyword Queries
Async SQL
Issue 6h Signed URL
3. AI & External Services
GenAI APIs
• Google Gemini (SDG Mapping)
• OpenAI (CV Review)
Google News RSS
4. Cloud Storage (Supabase)
PostgreSQL (17 Tables + RLS)
Private Storage (Lesson Media)
Direct Media Stream